Optimization Problem Statements and Mathematical Formulations

1. Diamond Drill Purchasing Optimization

A company manufactures three types of diamond drills: heavy-duty, medium-duty, and light-duty. The company purchases industrial-grade diamonds from three suppliers whose diamond mixes differ in quality. Supplier 1 charges \$3.00 per kilogram; 40% of its diamonds are suitable for heavy-duty drills, 35% for medium-duty drills, and 25% for light-duty drills. Supplier 2 charges \$2.50 per kilogram; 35% of its diamonds are suitable for heavy-duty drills, 55% for medium-duty drills, and 10% for light-duty drills. Supplier 3 charges \$2.70 per kilogram; 30% of its diamonds are suitable for heavy-duty drills, 30% for medium-duty drills, and 40% for light-duty drills. The company requires 1,700 kg of diamonds suitable for heavy-duty drills, 1,200 kg for medium-duty drills, and 1,800 kg for light-duty drills. The company must determine how many kilograms of diamonds to purchase from each supplier so that all three production requirements are satisfied at the minimum possible total purchasing cost.

2. Mathematical Formulation

Decision variables

x_1 = kilograms of diamonds purchased from Supplier 1
x_2 = kilograms of diamonds purchased from Supplier 2
x_3 = kilograms of diamonds purchased from Supplier 3

Objective function

Minimize total purchasing cost:

$$\text{Min } Z = 3.00x_1 + 2.50x_2 + 2.70x_3$$

Constraints

Heavy-duty diamond requirement:

$$0.40x_1 + 0.35x_2 + 0.30x_3 \geq 1700$$

Medium-duty diamond requirement:

$$0.35x_1 + 0.55x_2 + 0.30x_3 \geq 1200$$

Light-duty diamond requirement:

$$0.25x_1 + 0.10x_2 + 0.40x_3 \geq 1800$$

Non-negativity:

$$x_1, x_2, x_3 \geq 0$$